US012418633B2

(12) **United States Patent**
Turiello et al.

(10) **Patent No.: US 12,418,633 B2**
(45) **Date of Patent: Sep. 16, 2025**

(54) **METHODS AND SYSTEM OF INCIDENT BASED CAMERA DEVICE ACTIVATION IN A FIREFIGHTER AIR REPLENISHMENT SYSTEM HAVING BREATHABLE AIR SUPPLIED THEREIN**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 181 days.

(21) Appl. No.: **18/207,144**

(22) Filed: **Jun. 8, 2023**

(65) **Prior Publication Data**

US 2023/0319241 A1 Oct. 5, 2023

Related U.S. Application Data

(60) Provisional application No. 63/413,616, filed on Oct. 6, 2022, provisional application No. 63/359,882, filed (Continued)

(51) **Int. Cl.**
G08B 21/22 (2006.01)
A62B 27/00 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC ***H04N 7/188*** (2013.01); ***A62B 27/00*** (2013.01); *G16Y 40/50* (2020.01)

(58) **Field of Classification Search**
USPC 340/540, 545.2, 530, 539.1, 435, 474, 340/488, 539.22, 568.1
See application file for complete search history.

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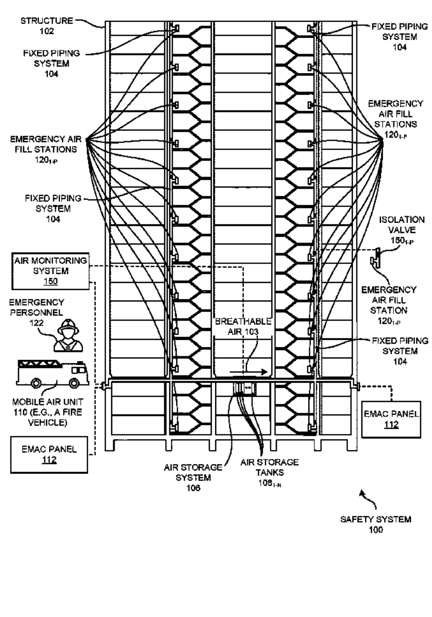
(57) **ABSTRACT**

Disclosed are methods and/or a system of incident based camera device activation in a safety system of a structure having a fixed piping system implemented therein to supply breathable air thereacross. In accordance therewith, one or more sensor(s) associated with one or more component(s) of the safety system is integrated with a computing platform executing on a data processing device. Based on the integration of the one or more sensor(s) with the computing platform, one or more environmental parameter(s) of the one or more component(s) of the safety system is sensed. One or more camera device(s) in a vicinity of and/or on the one or more component(s) of the safety system is automatically activated based on determining, from the sensing, occurrence of an incident.

20 Claims, 9 Drawing Sheets



Related U.S. Application Data

on Jul. 11, 2022, provisional application No. 63/357,743, filed on Jul. 1, 2022, provisional application No. 63/357,754, filed on Jul. 1, 2022, provisional application No. 63/356,996, filed on Jun. 29, 2022.

(51) **Int. Cl.**
H04N 7/18 (2006.01)
G16Y 40/50 (2020.01)

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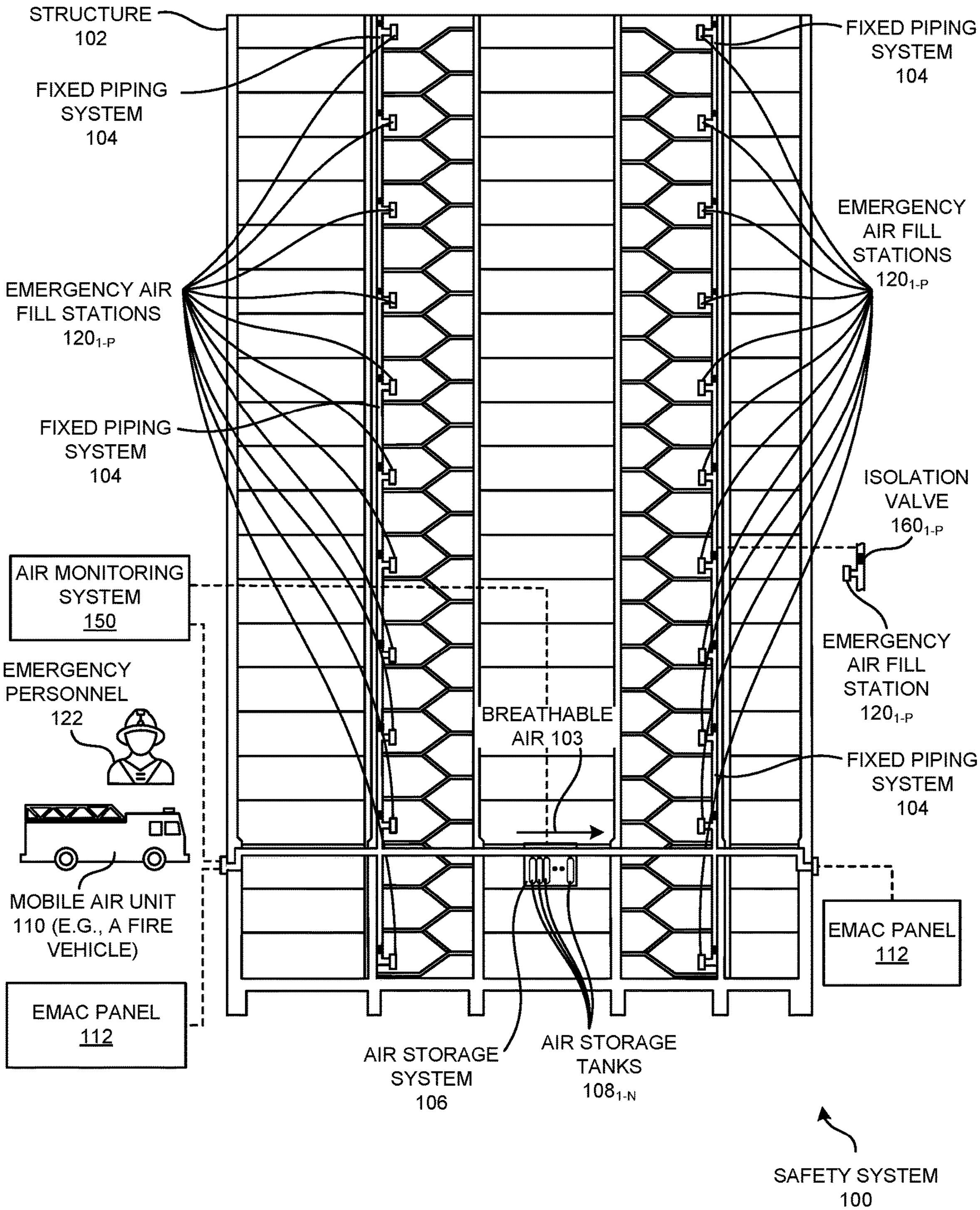
STRUCTURE 102
FIXED PIPING SYSTEM 104
FIXED PIPING SYSTEM 104
EMERGENCY AIR FILL STATIONS 120_{1-P}
EMERGENCY AIR FILL STATIONS 120_{1-P}
FIXED PIPING SYSTEM 104
ISOLATION VALVE 160_{1-P}
AIR MONITORING SYSTEM 150
EMERGENCY AIR FILL STATION 120_{1-P}
EMERGENCY PERSONNEL 122
BREATHABLE AIR 103
FIXED PIPING SYSTEM 104
MOBILE AIR UNIT 110 (E.G., A FIRE VEHICLE)
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EMAC PANEL 112
AIR STORAGE SYSTEM 106
AIR STORAGE TANKS 108_{1-N}
SAFETY SYSTEM 100

FIG. 1

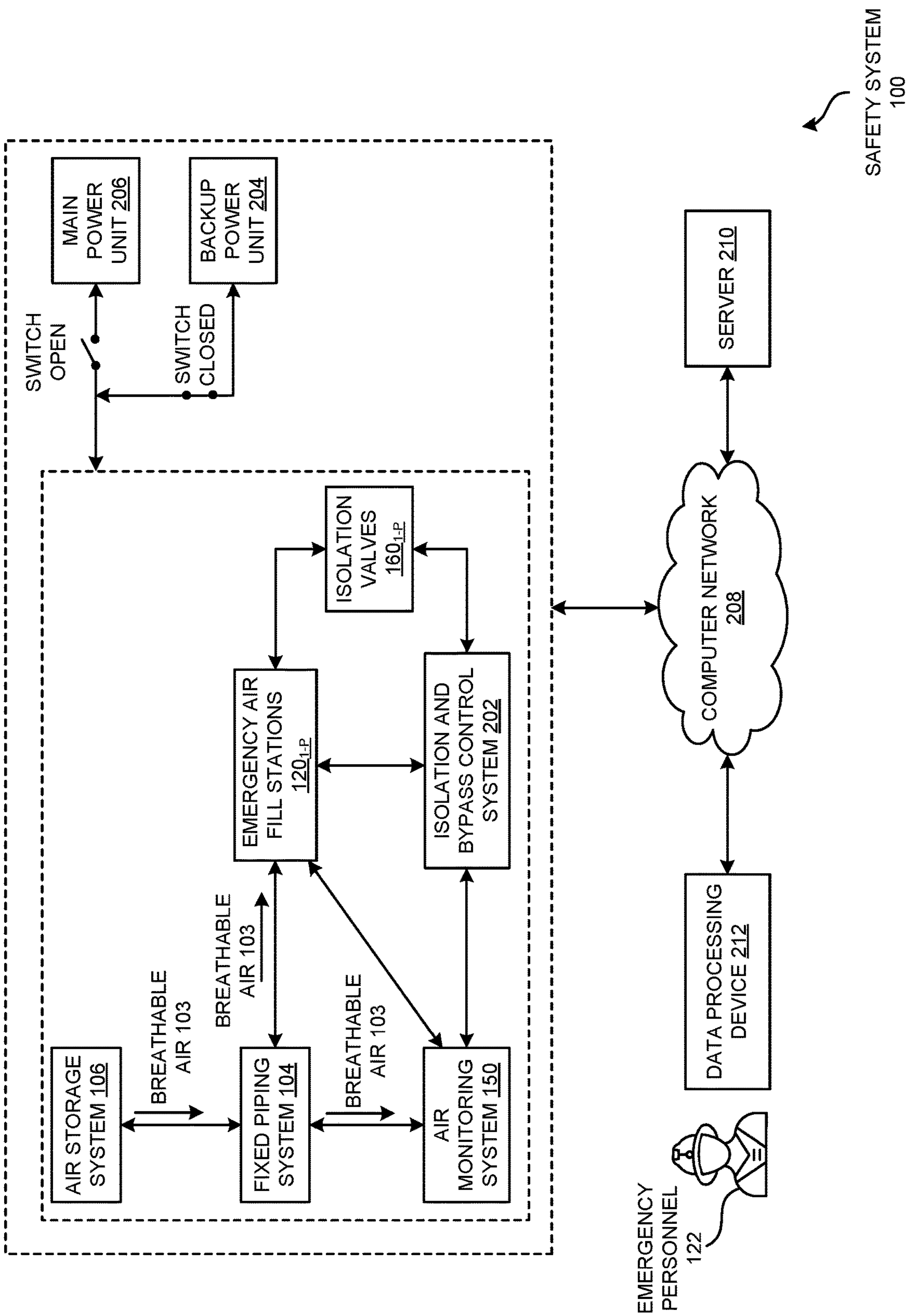

SAFETY SYSTEM 100
MAIN POWER UNIT 206
BACKUP POWER UNIT 204
SWITCH OPEN
SWITCH CLOSED
ISOLATION VALVES 160$_{1\text{-}P}$
EMERGENCY AIR FILL STATIONS 120$_{1\text{-}P}$
ISOLATION AND BYPASS CONTROL SYSTEM 202
AIR STORAGE SYSTEM 106
BREATHABLE AIR 103
FIXED PIPING SYSTEM 104
BREATHABLE AIR 103
BREATHABLE AIR 103
AIR MONITORING SYSTEM 150
SERVER 210
COMPUTER NETWORK 208
DATA PROCESSING DEVICE 212
EMERGENCY PERSONNEL 122

FIG. 2

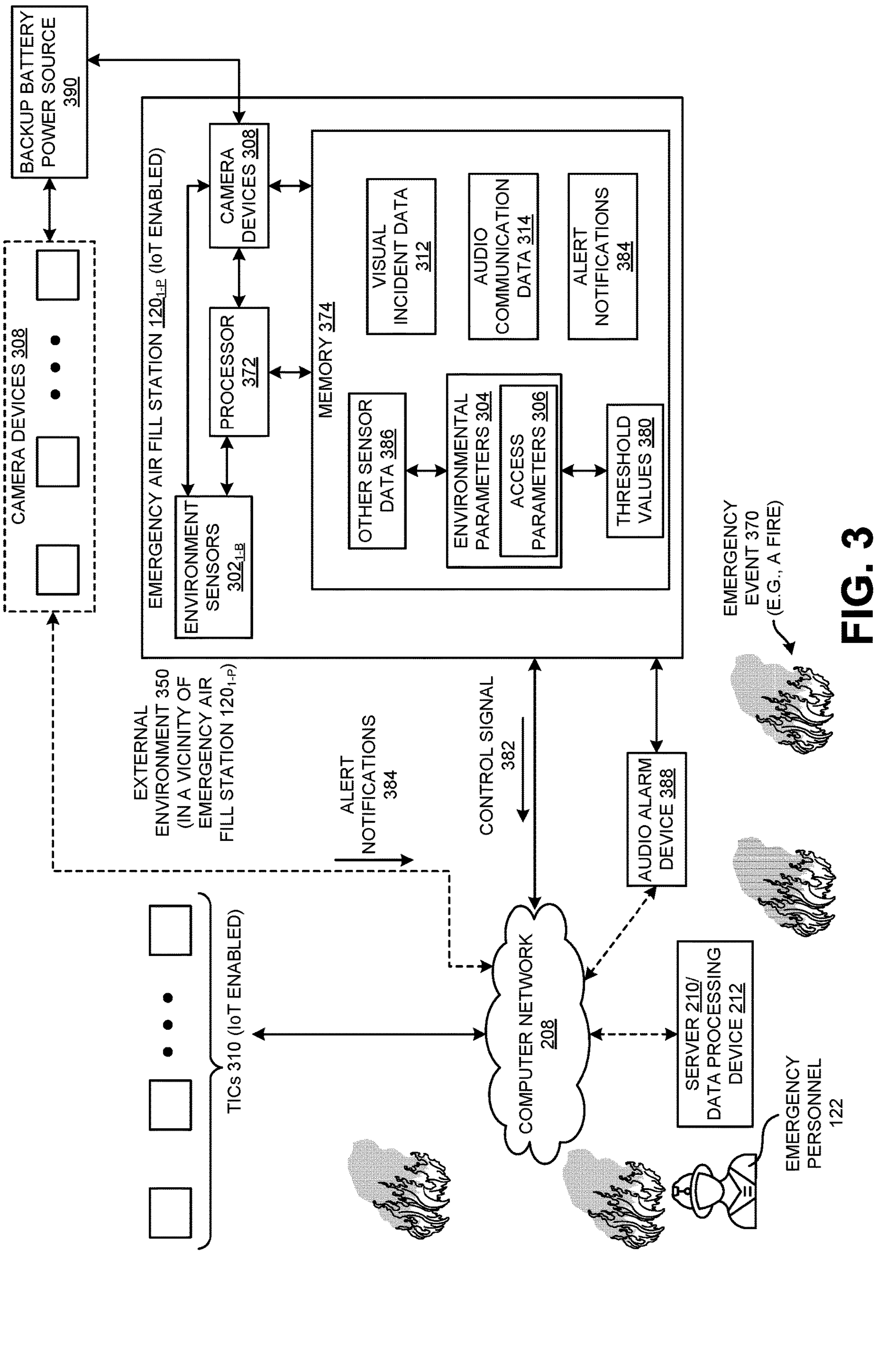
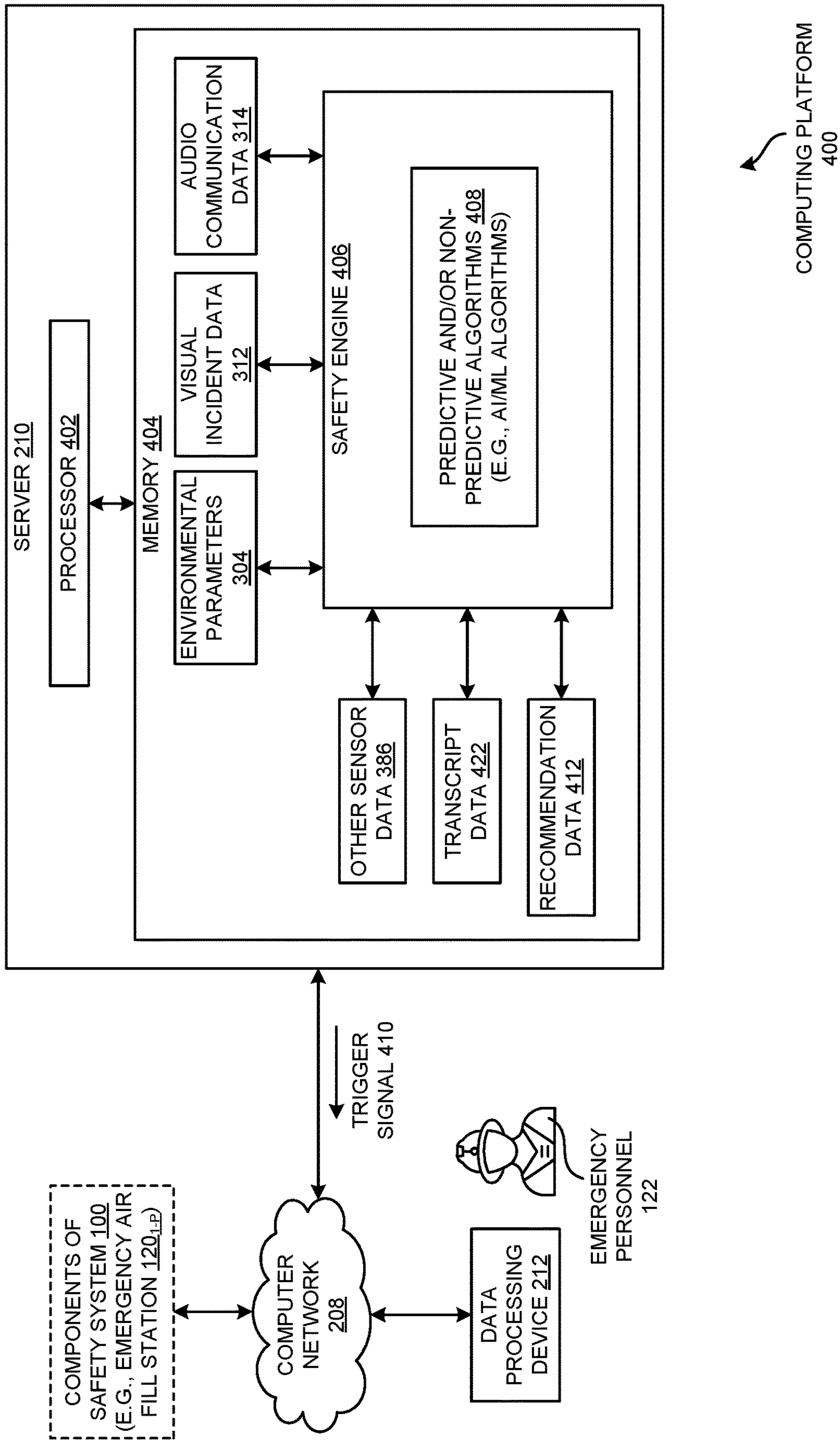

BACKUP BATTERY POWER SOURCE 390
CAMERA DEVICES 308
EMERGENCY AIR FILL STATION 120_{1-P} (IoT ENABLED)
CAMERA DEVICES 308
PROCESSOR 372
ENVIRONMENT SENSORS 302_{1-B}
MEMORY 374
VISUAL INCIDENT DATA 312
AUDIO COMMUNICATION DATA 314
ALERT NOTIFICATIONS 384
OTHER SENSOR DATA 386
ENVIRONMENTAL PARAMETERS 304
ACCESS PARAMETERS 306
THRESHOLD VALUES 380
EXTERNAL ENVIRONMENT 350 (IN A VICINITY OF EMERGENCY AIR FILL STATION 120_{1-P})
ALERT NOTIFICATIONS 384
CONTROL SIGNAL 382
AUDIO ALARM DEVICE 388
EMERGENCY EVENT 370 (E.G., A FIRE)
TICs 310 (IoT ENABLED)
COMPUTER NETWORK 208
SERVER 210/ DATA PROCESSING DEVICE 212
EMERGENCY PERSONNEL 122

FIG. 3



SERVER 210
PROCESSOR 402
MEMORY 404
ENVIRONMENTAL PARAMETERS 304
VISUAL INCIDENT DATA 312
AUDIO COMMUNICATION DATA 314
SAFETY ENGINE 406
PREDICTIVE AND/OR NON-PREDICTIVE ALGORITHMS 408 (E.G., AI/ML ALGORITHMS)
OTHER SENSOR DATA 386
TRANSCRIPT DATA 422
RECOMMENDATION DATA 412
COMPUTING PLATFORM 400
TRIGGER SIGNAL 410
COMPONENTS OF SAFETY SYSTEM 100 (E.G., EMERGENCY AIR FILL STATION 120_{1-P})
COMPUTER NETWORK 208
DATA PROCESSING DEVICE 212
EMERGENCY PERSONNEL 122

FIG. 4

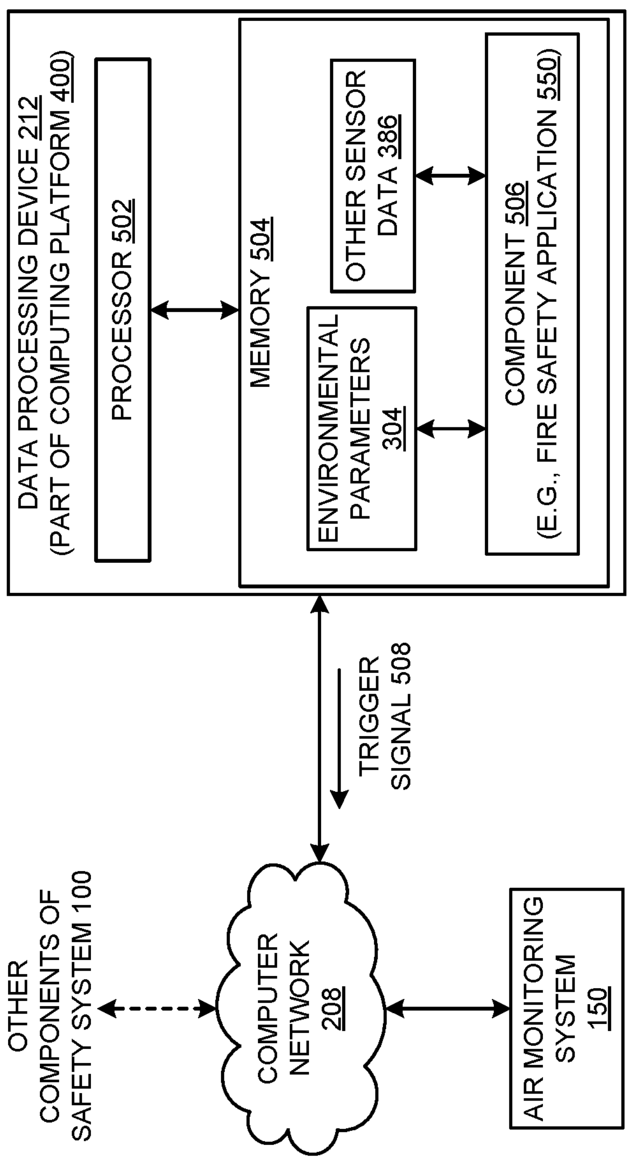

DATA PROCESSING DEVICE 212
(PART OF COMPUTING PLATFORM 400)
PROCESSOR 502
MEMORY 504
ENVIRONMENTAL PARAMETERS 304
OTHER SENSOR DATA 386
COMPONENT 506
(E.G., FIRE SAFETY APPLICATION 550)
TRIGGER SIGNAL 508
OTHER COMPONENTS OF SAFETY SYSTEM 100
COMPUTER NETWORK 208
AIR MONITORING SYSTEM 150

FIG. 5

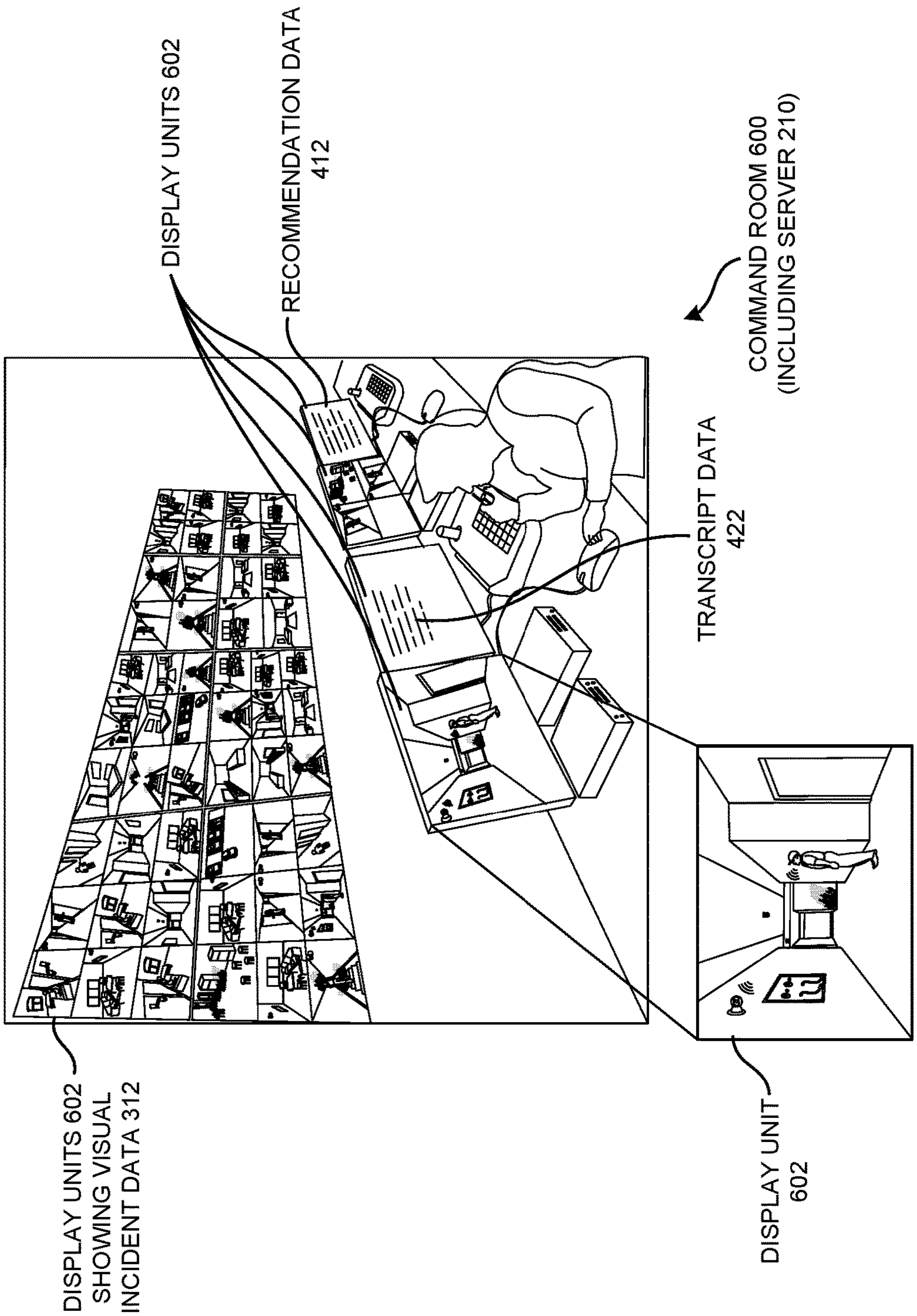

DISPLAY UNITS 602
RECOMMENDATION DATA 412
COMMAND ROOM 600 (INCLUDING SERVER 210)
TRANSCRIPT DATA 422
DISPLAY UNITS 602 SHOWING VISUAL INCIDENT DATA 312
DISPLAY UNIT 602

FIG. 6

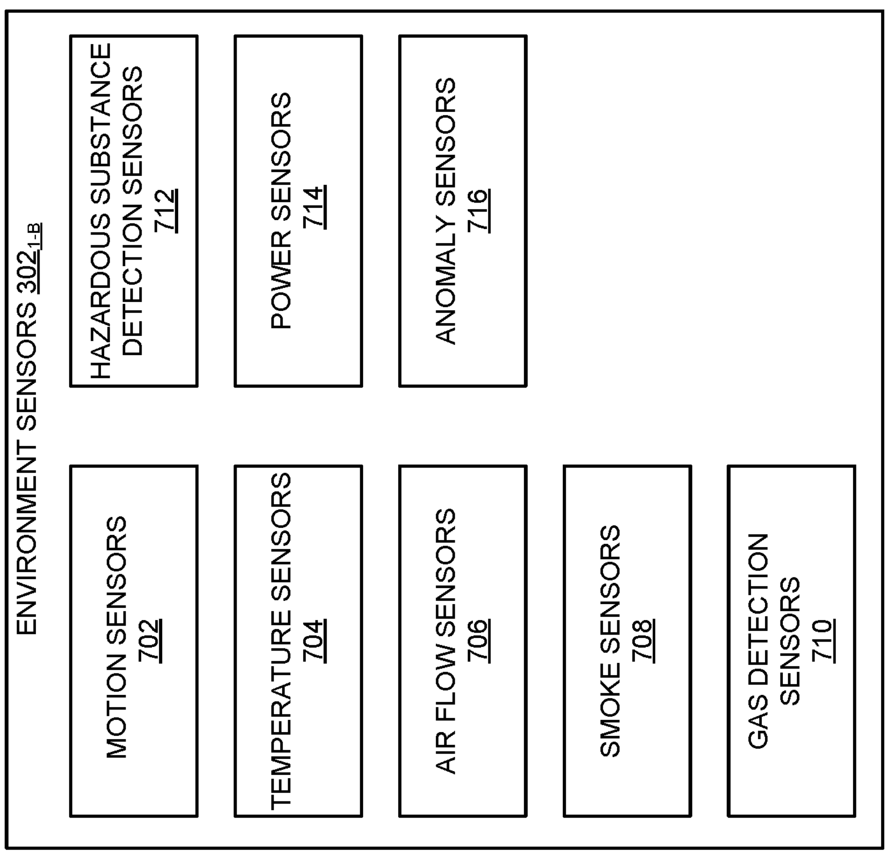

ENVIRONMENT SENSORS 302$_{1-B}$
MOTION SENSORS 702
TEMPERATURE SENSORS 704
AIR FLOW SENSORS 706
SMOKE SENSORS 708
GAS DETECTION SENSORS 710
HAZARDOUS SUBSTANCE DETECTION SENSORS 712
POWER SENSORS 714
ANOMALY SENSORS 716

FIG. 7

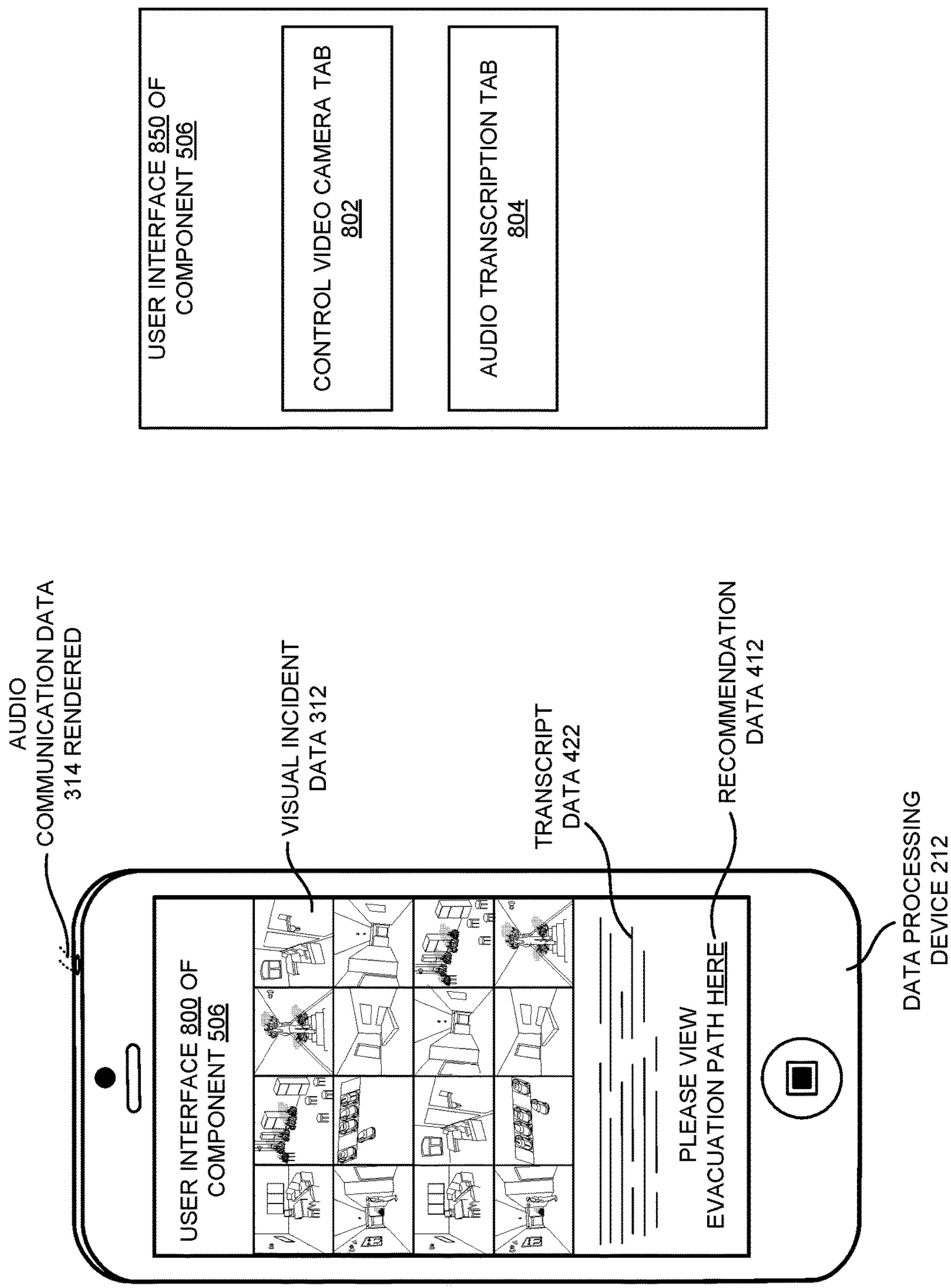

USER INTERFACE 850 OF COMPONENT 506
CONTROL VIDEO CAMERA TAB 802
AUDIO TRANSCRIPTION TAB 804
AUDIO COMMUNICATION DATA 314 RENDERED
VISUAL INCIDENT DATA 312
TRANSCRIPT DATA 422
RECOMMENDATION DATA 412
USER INTERFACE 800 OF COMPONENT 506
PLEASE VIEW EVACUATION PATH HERE
DATA PROCESSING DEVICE 212

FIG. 8

902

INTEGRATE ONE OR MORE SENSOR(S) ASSOCIATED WITH ONE OR MORE COMPONENT(S) OF A SAFETY SYSTEM OF A STRUCTURE WITH A COMPUTING PLATFORM EXECUTING ON A DATA PROCESSING DEVICE, WITH THE ONE OR MORE COMPONENT(S) RELATED TO ACCESS OF BREATHABLE AIR SUPPLIED ACROSS THE SAFETY SYSTEM VIA A FIXED PIPING SYSTEM IMPLEMENTED THEREIN

904

IN ACCORDANCE WITH THE INTEGRATION OF THE ONE OR MORE SENSOR(S) WITH THE COMPUTING PLATFORM,

SENSE ONE OR MORE ENVIRONMENTAL PARAMETER(S) OF THE ONE OR MORE COMPONENT(S) OF THE SAFETY SYSTEM, AND

AUTOMATICALLY ACTIVATE ONE OR MORE CAMERA DEVICE(S) IN A VICINITY OF AND/OR ON THE ONE OR MORE COMPONENT(S) OF THE SAFETY SYSTEM BASED ON DETERMINING, FROM THE SENSING OF THE ONE OR MORE ENVIRONMENTAL PARAMETER(S), OCCURRENCE OF AN INCIDENT

FIG. 9

METHODS AND SYSTEM OF INCIDENT BASED CAMERA DEVICE ACTIVATION IN A FIREFIGHTER AIR REPLENISHMENT SYSTEM HAVING BREATHABLE AIR SUPPLIED THEREIN

CLAIM OF PRIORITY

This application is a conversion application of, and claims priority to, U.S. Provisional Patent Application No. 63/356,996 titled CLOUD-BASED FIREFIGHTING AIR REPLENISHMENT MONITORING SYSTEM, SENSORS AND METHODS filed on Jun. 29, 2022, U.S. Provisional Patent Application No. 63/413,616 titled VIDEO CAMERA AT EMERGENCY AIR FILL PANEL FOR INCIDENT COMMAND VISUAL AND TRANSCRIPTION OF AUDIO VIA MOBILE DEVICE filed on Oct. 6, 2022, U.S. Provisional Patent Application No. 63/357,743 titled CONTINUAL AIR QUALITY MONITORING THROUGH LOCALIZED ANALYSIS OF BREATHABLE AIR THROUGH A SENSOR ARRAY filed on Jul. 1, 2022, U.S. Provisional Patent Application No. 63/357,754 titled ON-DEMAND CERTIFICATION THROUGH COMMUNICATION OF ASSOCIATED AIR-QUALITY MARKER DATA TO A REMOTE CERTIFICATION LABORATORY filed on Jul. 1, 2022, and U.S. Provisional Patent Application No. 63/359,882 titled REMOTE MONITORING AND CONTROL OF A FIREFIGHTER AIR REPLENISHMENT SYSTEM THROUGH SENSORS DISTRIBUTED WITHIN COMPONENTS OF THE FIREFIGHTER AIR REPLENISHMENT SYSTEM filed on Jul. 11, 2022. The contents of each of the aforementioned applications are incorporated herein by reference in entirety thereof.

FIELD OF TECHNOLOGY

This disclosure relates generally to emergency systems and, more particularly, to methods and/or a system of incident based camera device activation in a safety system of a structure having breathable air supplied therein.

BACKGROUND

A structure (e.g., a vertical building, a horizontal building, a tunnel, marine craft) may have a Firefighter Air Replenishment System (FARS) implemented therein. The FARS may have an emergency air fill station therein to enable firefighters and/or emergency personnel access breathable air therethrough. The FARS may have other components relevant to critical functioning thereof. An incident (e.g., a fire, smoke/air pollution) occurring in the structure in a vicinity of one or more components of the FARS may endanger lives of the emergency personnel and/or people within the structure. Reducing chances of occurrence of the incident may warrant repeated monitoring of the FARS. Despite the careful monitoring, the incident may recur. Even if occurrence of the incident is controlled through painstaking design of the FARS based on the careful monitoring, another emergency situation resulting in casualties and/or damage to the structure may occur.

SUMMARY

Disclosed are methods and/or a system of incident based camera device activation in a safety system of a structure having breathable air supplied therein.

In one aspect, a method of a safety system of a structure having a fixed piping system implemented therein to supply breathable air thereacross is disclosed. The method includes integrating one or more sensor(s) associated with one or more component(s) of the safety system with a computing platform executing on a data processing device. The one or more component(s) relates to access of the breathable air within the safety system. The method also includes, in accordance with the integration of the one or more sensor(s) with the computing platform, sensing one or more environmental parameter(s) of the one or more component(s) of the safety system, and automatically activating one or more camera device(s) in a vicinity of and/or on the one or more component(s) of the safety system based on determining, from the sensing of the one or more environmental parameter(s), occurrence of an incident.

In another aspect, a safety system of a structure having a fixed piping system implemented therein to supply breathable air thereacross is disclosed. The safety system includes one or more component(s) related to access of the breathable air within the safety system. one or more sensor(s) associated with the one or more component(s), and a data processing device executing a computing platform thereon to integrate the one or more sensor(s) with the computing platform. In accordance with the integration of the one or more sensor(s) with the computing platform, the one or more sensor(s) senses one or more environmental parameter(s) of the one or more component(s), and a processor associated with the one or more sensor(s) automatically activates one or more camera device(s) in a vicinity of and/or on the one or more component(s) based on determining, from the sensing of the one or more environmental parameter(s), occurrence of an incident.

In yet another aspect, a method of a safety system of a structure having a fixed piping system implemented therein to supply breathable air thereacross is disclosed. The method includes integrating one or more sensor(s) associated with one or more component(s) of the safety system with a computing platform executing on a data processing device. The one or more component(s) relates to access of the breathable air within the safety system. The method also includes, in accordance with the integration of the one or more sensor(s) with the computing platform, sensing one or more environmental parameter(s) of the one or more component(s) of the safety system, and automatically activating one or more camera device(s) in a vicinity of and/or on the one or more component(s) of the safety system based on determining, from the sensing of the one or more environmental parameter(s), occurrence of an incident. Further, the method includes, in accordance with the automatic activation of the one or more camera device(s), capturing visual data and/or audio data of the incident.

Other features will be apparent from the accompanying drawings and from the detailed description that follows.

BRIEF DESCRIPTION OF THE DRAWINGS

The embodiments of this invention are illustrated by way of example and not limitation in the figures of the accompanying drawings, in which like references indicate similar elements and in which:

FIG. **1** is a schematic and an illustrative view of a safety system associated with a structure, according to one or more embodiments.

FIG. **2** is a schematic view of the safety system of FIG. **1** with elements thereof integrated therewithin in detail, according to one or more embodiments.

FIG. 3 is a schematic view of an emergency air fill station of the safety system of FIGS. 1-2 and an illustrative view of a context of an emergency event in an external environment thereof, according to one or more embodiments.

FIG. 4 is a schematic view of a computing platform relevant to the safety system of FIGS. 1-2 implemented through a server, according to one or more embodiments.

FIG. 5 is a schematic view of a data processing device of FIGS. 2-4, according to one or more embodiments.

FIG. 6 is an illustrative view of a command room of the safety system of FIGS. 1-2.

FIG. 7 is a schematic view of example environment sensors of the emergency air fill station of FIG. 3.

FIG. 8 is an example user interface view of a component of the data processing device of FIGS. 2-5.

FIG. 9 is a process flow diagram detailing the operations involved in incident based camera device activation in a safety system of a structure having breathable air supplied therein, according to one or more embodiments.

Other features of the present embodiments will be apparent from the accompanying drawings and from the detailed description that follows.

DETAILED DESCRIPTION

Example embodiments, as described below, may be used to provide methods and/or a system of incident based camera device activation in a safety system of a structure having breathable air supplied therein. Although the present embodiments have been described with reference to specific example embodiments, it will be evident that various modifications and changes may be made to these embodiments without departing from the broader spirit and scope of the various embodiments.

FIG. 1 shows a safety system **100** associated with a structure **102**, according to one or more embodiments. In one or more embodiments, safety system **100** may be a Firefighter Air Replenishment System (FARS) to enable firefighters entering structure **102** in times of fire-related emergencies to gain access to breathable (e.g., human breathable) air (e.g., breathable air **103**) in-house without the need of bringing in air bottles/cylinders to be transported up several flights of stairs of structure **102** or deep thereinto, or to refill depleted air bottles/cylinders that are brought into structure **102**. In one or more embodiments, safety system **100** may supply breathable air provided from a supply of air tanks (to be discussed) stored in structure **102**. When a fire department vehicle arrives at structure **102** during an emergency, breathable air supply typically may be provided through a source of air connected to said vehicle. In one or more embodiments, safety system **100** may enable firefighters to refill air bottles/cylinders thereof at emergency air fill stations (to be discussed) located throughout structure **102**. Specifically, in some embodiments, firefighters may be able to fill air bottles/cylinders thereof at emergency air fill stations within structure **102** under full respiration in less than one to two minutes.

In one or more embodiments, structure **102** may encompass vertical building structures, horizontal building structures (e.g., shopping malls, hypermarts, extended shopping, storage and/or warehousing related structures), tunnels, marine craft (e.g., large marine vessels such as cruise ships, cargo ships, submarines and large naval craft, which may be "floating" versions of buildings and horizontal structures) and mines. Other structures are within the scope of the exemplary embodiments discussed herein. In one or more embodiments, safety system **100** may include a fixed piping system **104** permanently installed within structure **102** serving as a constant source of replenishment of breathable air **103**. Fixed piping system **104** may be regarded as being analogous to a water piping system within structure **102** or another structure analogous thereto for the sake of imaginative convenience.

As shown in FIG. 1, fixed piping system **104** may distribute breathable air **103** across floors/levels of structure **102**. For the aforementioned purpose, fixed piping system **104** may distribute breathable air **103** from an air storage system **106** (e.g., within structure **102**) including a number of air storage tanks **1081**-N that serve as sources of pressurized/compressed air (e.g., breathable air **103**). Additionally, in one or more embodiments, fixed piping system **104** may interconnect with a mobile air unit **110** (e.g., a fire vehicle) through an External Mobile Air Connection (EMAC) panel **112**.

In one or more embodiments, EMAC panel **112** may be a boxed structure (e.g., exterior to structure **102**) to enable the interconnection between mobile air unit **110** and safety system **100**. For example, mobile air unit **110** may include an on-board air compressor to store and replenish pressurized/compressed air (e.g., breathable air analogous to breathable air **103**) in air bottles/cylinders (e.g., utilizable with Self-Contained Breathing Apparatuses (SCBAs) carried by firefighters). Mobile air unit **110** may also include other pieces of air supply/distribution equipment (e.g., piping and/or air cylinders/bottles) that may be able to leverage the sources of breathable air **103** within safety system **100** through EMAC panel **112**. Firefighters, for example, may be able to fill breathable air (e.g., breathable air **103**, breathable air analogous to breathable air **103**) into air bottles/cylinders (e.g., spare bottles, bottles requiring replenishment of breathable air) carried on mobile air unit **110** through safety system **100**.

In FIG. 1, EMAC panel **112** is shown at two locations merely for the sake of illustrative convenience. In one or more embodiments, an air monitoring system **150** may be installed as part of safety system **100** to automatically track and monitor a parameter (e.g., pressure) and/or a quality (e.g., indicated by moisture levels, carbon monoxide levels) of breathable air **103** within safety system **100**. FIG. 1 shows air monitoring system **150** as communicatively coupled to air storage system **106** and EMAC panel **112** merely for the sake of example. It should be noted that EMAC panel **112** may be at a remote location associated with (e.g., internal to, external to) structure **102**. In one or more embodiments, for monitoring the parameters and/or the quality of breathable air within safety system **100**, air monitoring system **150** include appropriate sensors and circuitries therein. For example, a pressure sensor (to be discussed) within air monitoring system **150** may automatically sense and record a pressure of breathable air **103** of safety system **100**. Said pressure sensor may communicate with an alarm system that is triggered when the sensed pressure is outside a safety range. Also, in one or more embodiments, air monitoring system **150** may automatically trigger a shutdown of breathable air distribution through safety system **100** in case of impurity/contaminant (e.g., carbon monoxide) detection therethrough yielding levels above a safety/predetermined threshold.

In one or more embodiments, fixed piping system **104** may include pipes (e.g., constituted out of stainless steel tubing) that distribute breathable air **103** to a number of emergency air fill stations $\mathbf{120}_{1\text{-}P}$ within structure **102**. In one example implementation, each emergency air fill station $\mathbf{120}_{1\text{-}P}$ may be located at a specific level of structure **102**. If structure **102** is regarded as a vertical building structure, an emergency air fill station $\mathbf{120}_{1\text{-}P}$ may be located at each of a basement level, a first floor level, a second floor level and so on. For example, emergency air fill station $\mathbf{120}_{1\text{-}P}$ may be located at the end of the flight of stairs that emergency fighting personnel (e.g., firefighting personnel) need to climb to reach a specific floor level within the vertical building structure.

In one or more embodiments, an emergency air fill station $\mathbf{120}_{1\text{-}P}$ may be a static location within a level of structure **102** that provides emergency personnel **122** (e.g., firefighters, emergency responders) with the ability to rapidly fill air bottles/cylinders (e.g., SCBA cylinders) with breathable air **103**. In one or more embodiments, emergency air fill station $\mathbf{120}_{1\text{-}P}$ may be an emergency air fill panel or a rupture containment air fill station. In one or more embodiments, proximate each emergency air fill station $\mathbf{120}_{1\text{-}P}$, safety system **100** may include an isolation valve $\mathbf{160}_{1\text{-}P}$ to isolate a corresponding emergency air fill station $\mathbf{120}_{1\text{-}P}$ from a rest of safety system **100**. For example, said isolation may be achieved through the manual turning of isolation valve $\mathbf{160}_{1\text{-}P}$ proximate the corresponding emergency air fill station $\mathbf{120}_{1\text{-}P}$ or remotely (e.g., based on automatic turning) from air monitoring system **150**. In one example implementation, air monitoring system **150** may maintain breathable air supply to a subset of emergency air fill stations $\mathbf{120}_{1\text{-}P}$ via fixed piping system **104** through control of a corresponding subset of isolation valves $\mathbf{160}_{1\text{-}P}$ and may isolate the other emergency air fill stations $\mathbf{120}_{1\text{-}P}$ from the breathable air supply. It should be noted that configurations and components of safety system **100** may vary from the example safety system **100** of FIG. **1**.

FIG. **2** shows safety system **100** with elements thereof integrated therewithin in detail, according to one or more embodiments. In one or more embodiments, safety system **100** may include air monitoring system **150** discussed above communicatively coupled to fixed piping system **104**, to which emergency air fill stations $\mathbf{120}_{1\text{-}P}$ are also coupled. In one or more embodiments, as seen above, the source of breathable air **103** may be air storage system **106**. In one or more embodiments, safety system **100** may also include an isolation and bypass control system **202** that is constituted by a set of electrical, mechanical and/or electronic components working together to automatically include and/or bypass one or more emergency air fill station(s) $\mathbf{120}_{1\text{-}P}$. For the aforementioned purpose, in one or more embodiments, isolation valve(s) $\mathbf{160}_{1\text{-}P}$ associated with the aforementioned emergency air fill stations $\mathbf{120}_{1\text{-}P}$ may be controlled (e.g., by opening or closing one or more of said isolation valves $\mathbf{160}_{1\text{-}P}$) by isolation and bypass control system **202**.

Further, in one or more embodiments, safety system **100** may include a backup power unit **204** (e.g., an electrical power system with electronic integration) to ensure uninterrupted power to components of safety system **100** during emergencies (e.g., a power cut, a mains power issue, a fire accident effected power issue). For the aforementioned purpose, in one or more embodiments, backup power unit **204** may be switched on in the case of a power related emergency with respect to a main power unit **206** (e.g., Alternating Current (AC) mains power, Direct Current (DC) power) associated with safety system **100**.

In one or more embodiments, one or more or all of the abovementioned components of safety system **100** may be integrated with sensor(s) to detect environmental conditions thereof. In one or more embodiments, based on the detection of the environmental conditions thereof, camera devices (e.g., video and/or audio; to be discussed below) may be automatically turned on to capture visuals and/or audio data of environments associated with the one or more components of safety system **100**. In one or more embodiments, the one or more components may be communicatively coupled through a computer network **208** (e.g., a Local Area Network (LAN), a Wide Area Network (WAN), a cloud computing network, a short-range communication network based on Bluetooth®, WiFi® and the like) to a remote server **210** (e.g., a network of servers, a single server, a distributed network of servers, a command room server associated with safety system **100** and so on). As will be discussed below, in one or more embodiments, server **210** may obtain data from the sensor(s), camera devices and other data from safety system **100**, perform analyses (e.g., predictive, non-predictive) thereof and provide recommendations (e.g., situational awareness based) based on the analyses.

In addition, in one or more embodiments, safety system **100** may include a data processing device **212** (e.g., a mobile phone, a tablet, an iPad®, a laptop, a desktop) also communicatively coupled to one or more components or each component of safety system **100** and server **210** through computer network **208**. Thus, in one or more embodiments, one or more components or each component of safety system **100** may have interfaces (not explicitly shown) for wireless communication through computer network **208**. Also, as will be discussed below, in one or more embodiments, wherever possible, elements (e.g., handheld Thermal Imaging Cameras (TICs), portable TICs, aerial TICs, camera devices, audio devices, light devices, one or more or all sensors discussed herein) may be Internet of Things (IoT) devices capable of collecting and feeding data to server **210** through computer network **208**. In one or more embodiments, IoT devices (or IoT enabled devices) may be devices and/or components with programmable hardware that can transmit data over computer networks (e.g., computer network **208** such as the Internet and/or other networks); said IoT devices may include or be associated with edge devices (not shown) to control data flow at the boundaries to computer network **208**.

FIG. **3** shows an emergency air fill station $\mathbf{120}_{1\text{-}P}$, according to one or more embodiments. Again, in one or more embodiments, emergency air fill station $\mathbf{120}_{1\text{-}P}$ may include one or more environment sensors $\mathbf{302}_{1\text{-}B}$ integrated therewith configured to sense environmental parameters **304** (e.g., temperature, audio alarm detection (e.g., a person screaming "fire!"), pressure, smoke, motion, ambient light) associated with an environment (e.g., external environment **350**) in an immediate vicinity of emergency air fill station $\mathbf{120}_{1\text{-}P}$. In one or more embodiments, environment sensors $\mathbf{302}_{1\text{-}B}$ may also sense access (e.g., access parameters **306** that are part of environmental parameters **304** in FIG. **3**) of and attempts to access emergency air fill station $\mathbf{120}_{1\text{-}P}$ by emergency personnel **122** (e.g., maintenance personnel, firefighters, emergency responders) and/or unauthorized personnel (e.g., example access by unauthorized personnel may involve tampering of one or more element(s) of emergency air fill station $\mathbf{120}_{1\text{-}P}$). In one or more embodiments, emergency air fill station $\mathbf{120}_{1\text{-}P}$ may have one or more camera devices **308** integrated therewith or external (e.g., in external environment **350**) thereto. In some embodiments, camera devices **308** may be considered as encompassing one or more environment sensors $\mathbf{302}_{1\text{-}B}$ (e.g., motion detection sensors); FIG. **3** shows camera devices **308** as distinct from environment sensors $\mathbf{302}_{1\text{-}B}$ merely for example purposes.

In one or more embodiments, emergency air fill station $\mathbf{120}_{1\text{-}P}$ may include a processor **372** (e.g., a microcontroller, a processor core, a single processor) communicatively coupled to a memory **374** (e.g., a volatile and/or a non-volatile memory). In one or more embodiments, environment sensors $\mathbf{302}_{1\text{-}B}$ may be interfaced with processor **372** and all of the abovementioned data/parameters (e.g., environmental parameters **304**) may be stored in memory **374**, as shown in FIG. **3**. FIG. **3** also shows TICs **310** as part of safety system **100** and in external environment **350** of emergency air fill station $\mathbf{120}_{1\text{-}P}$, according to one or more embodiments. In one or more embodiments, TICs **310** may be infrared cameras that sense infrared energy of objects to render images/video frames thereof corresponding to surface temperatures of said objects. In one or more embodiments, emergency personnel **122** may employ said TICs **310** to detect obstacles on the paths to/around emergency air fill stations $\mathbf{120}_{1\text{-}P}$ under low visibility; this may enable emergency personnel **122** perform rescue operations efficiently. As discussed and implied above, TICs **310** may be integrated with IoT capabilities to transmit data to server **210** through computer network **208**. Said data may be part of access parameters **306** or separate data transmitted to server **210**.

It should be noted that the sensing, detection and/or transmission of data to server **210** discussed above with regard to emergency air fill station $\mathbf{120}_{1\text{-}P}$ may also be performed at a device external to emergency air fill station $\mathbf{120}_{1\text{-}P}$. In such implementations, the external device itself may obviously be a component of safety system **100** with IoT/wireless communication capabilities. While FIG. **3** has been discussed with regard to an emergency air fill station $\mathbf{120}_{1\text{-}P}$, concepts discussed herein may be applicable across other components of safety system **100** such as air monitoring system **150**, air storage system **106**, isolation and bypass control system **202** and even backup power unit **204**.

FIG. **4** shows a computing platform **400** relevant to the FARS of safety system **100** implemented through server **210**, according to one or more embodiments. In one or more embodiments, server **210** may be a distributed (e.g., across a cloud) network of servers, a cluster of servers or a standalone server. As discussed above, in some embodiments, server **210** may be implemented as part of a fire command room within safety system **100**; additionally or alternatively, server **210** may be implemented external to safety system **100**. As shown in FIG. **4**, server **210** may include a processor **402** (e.g., a processor core, a network of processors, a single processor), communicatively coupled to a memory **404** (e.g., a volatile and/or a non-volatile memory). In one or more embodiments, memory **404** may include a safety engine **406** associated with the FARS stored therein and executable through processor **402**; safety engine **406** may integrate with environment sensors $\mathbf{302}_{1\text{-}B}$ (and all other sensors within safety system **100**) based on execution thereof through processor **402**.

FIG. **4** shows memory **404** as including data (e.g., detected, sensed; environmental parameters **304**) from one or more components of safety system **100**; the limited amount of data shown must not be considered as limiting the scope of the exemplary embodiments discussed herein. In one or more embodiments, safety engine **406** may have one or more predictive and/or non-predictive algorithms (e.g., predictive and/or non-predictive algorithms **408**) including Artificial Intelligence (AI)/Machine Learning (ML) based algorithms stored therein and executable through processor **402**.

In one or more embodiments, execution of predictive and/or non-predictive algorithms **408** through processor **402** may involve taking the abovementioned data and providing analyses and/or recommendations, as discussed above. It should be noted that each of the aforementioned data (e.g., environmental parameters **304**) may be real-time data from elements/components of safety system **100**. In one or more embodiments, analyses of the data and recommendations may result in increased situational awareness during emergencies/maintenance situations and improved efficiency with regard to safety system **100** and safety/security thereof.

In one or more implementations, the components (e.g., emergency air fill station $\mathbf{120}_{1\text{-}P}$, air storage system **106**, air monitoring system **150**) of safety system **100** may automatically transmit data (e.g., environmental parameters **304**) thereof to server **210**; server **210** may transmit trigger signals (e.g., trigger signal **410**) therefor. FIG. **5** shows data processing device **212** (e.g., a mobile phone, a tablet, a smart device, a laptop) in detail, according to one or more embodiments. In one or more embodiments, again, data processing device **212** may include a processor **502** (e.g., a single processor, a processor core) communicatively coupled to a memory **504** (e.g., a volatile and/or a non-volatile memory). In one or more embodiments, memory **504** may include a component **506** of safety engine **406** stored therein and enabled/provided through processor **402** of server **210**. FIG. **5** shows component **506** as a fire safety application **550** merely for example purposes. Again, in one or more embodiments, access to the data of one or more components of safety system **100** may be available to data processing device **212** via component **506** (e.g., through computer network **208** via safety engine **406** of server **210**). FIG. **5** also shows capabilities to control components of safety system **100** through data processing device **212** via trigger signals; FIG. **5** specifically shows a trigger signal **508** to initiate collection of data from air monitoring system **150** merely for example purposes. Again, in some implementations, data may be automatically communicated to data processing device **212** and in some others, data processing device **212** may trigger (e.g., through trigger signal **508**) collection thereof.

Referring back to FIG. **3**, each camera device **308** may be a programmable device to capture and record visual incidents and/or audio communications in external environment **350**. In some implementations, camera devices **308** may be integrated with TICs **310**; in some other implementations, camera devices **308** may be distinct from TICs **310**; further, in some other implementations, camera devices **308** may be the same as TICs **310**. In some embodiments, one or more camera devices **308** may include motion sensor(s) (e.g., example environment sensors $\mathbf{302}_{1\text{-}B}$) and/or facial recognition algorithms programmed therein to detect visual incidents such as tampering of emergency air fill station $\mathbf{120}_{1\text{-}P}$ (and, analogously, other components of safety system **100**).

FIG. **3** illustrates a fire as an example emergency event **370**. In this context, a temperature of external environment **350** may exceed a threshold value thereof. The ambient temperature of external environment **350** may be detected by one or more environment sensors $\mathbf{302}_{1\text{-}B}$. As part of determining/detecting emergency event **370**, processor **372** may determine that the temperature sensed through the one or more environment sensors $\mathbf{302}_{1\text{-}B}$ exceeds the threshold value thereof to automatically activate one or more camera devices **308** to capture visual incidents and/or audio communication in external environment **350** associated with emergency event **370**. FIG. **3** shows visual incident data **312** (e.g., images and/or video frames, a video sequence) and audio communication data **314** (e.g., audio accompanying visual incident data **312**, separate audio data) being stored in memory **374** based on the capturing thereof through the one or more camera devices **308**. It should be noted that memory **374** and/or processor **372** may even be part of the one or more camera devices **308**.

In another scenario, environment sensors $\mathbf{302}_{1\text{-}B}$ may include an audio level sensor to detect an ambient decibel level of audio/sound in external environment **350**. Here, emergency event **370** may involve emergency personnel **122** or a potential victim screaming "Fire!" The aforementioned scream may cause a decibel level of the ambient sound to exceed a threshold value thereof; processor **372** may determine that the ambient decibel level is in excess of the threshold value thereof to automatically activate the one or more camera devices **308** (and/or TICs **310**) discussed above to capture visual incident data **312** and audio communication data **314**. In more sophisticated implementations, processor **372** may execute algorithms to glean emergency event **370** from an interpretation of audio communication data **314** in real-time; alternatively or additionally, audio communication data **314** and/or visual incident data **312** may be transmitted to server **210** and server **210** may glean emergency event **370** based on executing safety engine **406** to remotely activate the one or more camera devices **308** discussed above. It should be noted that the same remote operation may be performed through data processing device **212** based on executing component **506**.

In one or more embodiments, emergency event **370** may include but is not limited to a fire hazard, an explosion, a smoke situation, a terrorist attack, tampering of one or more components of safety system **100**, air pollution in external environment **350**, increased hazardous components in breathable air **103**, and reduced pressure of breathable air **103**. In some implementations, emergency event **370** may even be a maintenance event or a simulated event (e.g., part of a demonstration of safety system **100** and/or one or more components thereof) based on triggering (e.g., through server **210**, data processing device **212**) environment sensors $\mathbf{302}_{1\text{-}B}$ to detect anomalous environmental parameters **304** and/or processor **372** appropriately. Thus, environment sensors $\mathbf{302}_{1\text{-}B}$ may also encompass internal pressure sensors configured to sense pressure of breathable air **103** and air component level sensors configured to sense levels of hazardous components of breathable air **103**. FIG. **3** shows threshold values **380** used by processor **372** to determine emergency event **370** based on comparison of environmental parameters **304** with threshold values **380**; based on the determination, processor **372** may automatically activate (e.g., based on transmitting a control signal **382** to the one or more camera devices **308**) the one or more camera devices **308** discussed above.

In one or more embodiments, camera devices **308** (and TICs **310**) may employ advanced night vision to capture visual incident data **312** during conditions of low visibility. In some implementations, one or more camera devices **308** may employ 360 degree pan-tilt-zoom (PTZ) features to enable emergency personnel **122** at server **210** and/or data processing device **212** to remotely control a movement and/or positioning (movement and/or positioning are merely two example camera device parameters) of the one or more camera devices **308** based on control signals therefor. Additionally, in one or more embodiments, the one or more camera devices **308** may transmit alert notifications (e.g., alert notifications **384** stored in memory **374**) to server **210** and/or data processing device **212** related to alerting server **210** and/or data processing device **212** (e.g., through component **506**) of emergency event **370**.

Referring back to FIG. **4**, server **210** may also store visual incident data **312** and audio communication data **314** in memory **404** for analyses thereof (to be discussed herein). For the aforementioned purpose, server **210** may also leverage cloud storage through computer network **208**. In one or more implementations, environment sensors $\mathbf{302}_{1\text{-}B}$ may be configured to detect environmental parameters **304** at all times and the one or more camera devices **308** discussed above may be activated solely during emergency event **370** to provide situational context to emergency personnel **122** at server **210** and/or data processing device **212** and/or personnel (e.g., authorized, unauthorized) within structure **100** in external environment **350**. In one or more embodiments, predictive and/or non-predictive algorithms **408** executing as part of safety engine **406** on server **210** may even take visual incident data **312** and/or audio communication data **314** to generate a transcript (e.g., transcript data **422**) thereof. Alternatively, transcript data **422** may be created based on leveraging cloud capabilities/services by server **210**. In some implementations, processor **372** may itself generate transcript data **422**.

As discussed above, in one or more embodiments, environmental parameters **304** may also be transmitted to server **210** and/or data processing device **212** for analysis thereat. In some implementations, predictive and/or non-predictive algorithms **408** executing on server **210** may analyze environmental parameters **304** and other sensor data **386** (in FIG. **3**; e.g., data collected by environment sensors $\mathbf{302}_{1\text{-}B}$) to provide device renderable recommendations (e.g., device renderable recommendation data **412** shown stored in memory **404** of server **210**). Recommendation data **412** may be associated with but may not be limited to preventive measures to control the fire discussed above as emergency event **370**, optimizing resources, directing emergency personnel **122** via data processing device **212** (or one or more audio/video devices (e.g., a public speaker system) within safety system **100**) across safety system **100** and generating an emergency map for effective evacuation of victims. Recommendation data **412** and/or transcript data **422**, in some implementations, may be generated at data processing device **212** based on execution of component **506** thereon.

FIG. **6** shows an example command room **600** implementation of server **210**. Here, server **210** may have a number of display units **602** associated therewith to view visual incident data **312** captured by the one or more camera devices **308** in real-time. In addition, one or more display units **602** may include audio rendering devices (not shown) thereon to render audio communication data **314** in real-time. Further, the one or more display units **602** may display transcript data **422** and/or recommendation data **412** discussed above. Referring back to FIG. **3**, during emergency event **370**, the one or more camera devices **308**, in conjunction with processor **372** and/or remote communication from server **210**/data processing device **212**, may activate an audio alarm device **388** (e.g., rendering pre-recorded sound, rendering an audio message) to apprise emergency personnel **122**/other personnel within structure **102**, at server **210**, at data processing device **212** and/or within command room **600** of emergency event **370**. In one or more embodiments, the one or more camera devices **308** discussed above may have a backup battery power source **390** associated therewith to supply interrupted power thereto during emergency event **370** (e.g., associated with power interruption).

FIG. **7** shows examples of environment sensors $\mathbf{302}_{1\text{-}B}$. As seen in FIG. **7**, environment sensors $\mathbf{302}_{1\text{-}B}$ may include but are not limited to motion sensors **702**, a temperature sensor **704**, air flow sensors **706**, smoke sensors **708**, gas detection sensors **710**, hazardous substance detection sensors **712**, power sensors **714** and anomaly sensors **716** (e.g., sensing malfunctioning of equipment). FIG. **8** shows visual incident data **312** and audio communication data **314** being rendered via component **506** (e.g., fire safety application **550**) executing on data processing device **212**. Again, transcript data **422** and/or recommendation data **412** may be rendered via a user interface **800** of component **506**. Emergency personnel **122** may control (e.g., through control video camera tab **802**) the one or more camera devices **308** discussed above and transcribe (e.g., using audio transcription tab **804**) audio communication data **314** via another user interface **850** of component **506**.

Thus, exemplary embodiments discussed herein may serve as an advance surveillance system implemented as part of safety system **100**. The capabilities discussed herein may enable safety system **100** to provide better situational awareness to emergency personnel **122** at server **210**, control room **600**, data processing device **122** and/or other personnel within structure **102**. Further, in one or more embodiments, safety system **100** discussed herein may provide for efficient contextual monitoring of safety system **100** and transmitting actionable recommendations viewable, hearable and/or readable by emergency personnel **122**/other personnel within structure **102**. It should be noted that all operations and/or functionalities discussed herein may be performed through one or processors (e.g., processor **372**, processor **402**, processor **502**) of one or more data processing devices (e.g., emergency air fill station $\mathbf{120}_{1\text{-}P}$, server **210**, data processing device **212**) of safety system **100** discussed above in conjunction with one or more other elements (e.g., environment sensors $\mathbf{302}_{1\text{-}B}$).

Also, it should be noted that both component **506** and safety engine **406** may be regarded as a computing platform analogous to computing platform **400** based on capabilities (e.g., including integration capabilities) provided thereto. Further, it should be noted that environment sensors $\mathbf{302}_{1\text{-}B}$ may not only sense parameters relevant to external environment **350** but also sense internal parameters relevant to emergency air fill station $\mathbf{120}_{1\text{-}P}$. The same discussion may analogously be applicable to other components of safety system **100** (e.g., air monitoring system **150**, air storage system **106**, isolation and bypass control system **202**, backup power unit **204**). Last but not the least, emergency event **370** discussed above may be generalized to detection of any incident (e.g., a real-time incident determined based on environmental parameters **304**). All reasonable variations are within the scope of the exemplary embodiments discussed herein.

FIG. **9** shows a process flow diagram detailing the operations involved in incident based camera device activation in a safety system (e.g., safety system **100**) of a structure (e.g., structure **102**) having breathable air (e.g., breathable air **103**) supplied therein via a fixed piping system (e.g., fixed piping system **104**), according to one or more embodiments. In one or more embodiments, operation **902** may involve integrating one or more sensor(s) (e.g., environment sensors $\mathbf{302}_{1\text{-}B}$) associated with one or more component(s) (e.g., emergency air fill station $\mathbf{120}_{1\text{-}P}$, air monitoring system **150**, air storage system **106**, isolation and bypass control system **202**) of the safety system with a computing platform (e.g., safety engine **406**, component **506**) executing on a data processing device (e.g., server **210**, data processing device **212**). In one or more embodiments, the one or more component(s) may relate to access of the breathable air within the safety system.

In one or more embodiments, operation **904** may then involve, in accordance with the integration of the one or more sensor(s) with the computing platform, sensing one or more environmental parameter(s) (e.g., environmental parameters **304**) of the one or more component(s) of the safety system, and automatically activating one or more camera device(s) (e.g., camera devices **308**) in a vicinity (e.g., in external environment **350**) of and/or on the one or more component(s) of the safety system based on determining, from the sensing of the one or more environmental parameter(s), occurrence of an incident (e.g., emergency event **370**).

Although the present embodiments have been described with reference to specific example embodiments, it will be evident that various modifications and changes may be made to these embodiments without departing from the broader spirit and scope of the various embodiments.

A number of embodiments have been described. Nevertheless, it will be understood that various modifications may be made without departing from the spirit and scope of the claimed invention. In addition, the logic flows depicted in the figures do not require the particular order shown, or sequential order, to achieve desirable results. In addition, other steps may be provided, or steps may be eliminated, from the described flows, and other components may be added to, or removed from, the described systems. Accordingly, other embodiments are within the scope of the following claims.

The structures and modules in the figures may be shown as distinct and communicating with only a few specific structures and not others. The structures may be merged with each other, may perform overlapping functions, and may communicate with other structures not shown to be connected in the figures. Accordingly, the specification and/or drawings may be regarded in an illustrative rather than a restrictive sense.

What is claimed is:

1. A method of a safety system of a structure having a fixed piping system implemented therein to supply breathable air thereacross, comprising:

integrating at least one sensor associated with at least one component of the safety system with a computing platform executing on a data processing device, the at least one component related to access of the breathable air within the safety system; and

in accordance with the integration of the at least one sensor with the computing platform,

sensing at least one environmental parameter of the at least one component of the safety system, wherein the at least one environmental parameter includes pressure, smoke, motion, audio detection, temperature, or ambient light;

determining, from the sensing of the at least one environmental parameter, an occurrence of an incident corresponding to the at least one environmental parameter; and

automatically activating at least one camera device in a vicinity of the at least one environmental parameter and on the at least one component of the safety system responsive to the occurrence of the incident.

2. The method of claim **1**, further comprising capturing, through the automatically activated at least one camera device, at least one of: visual data and audio data of the incident.

3. The method of claim **2**, further comprising generating, through at least one of: the data processing device and another data processing device communicatively coupled to the data processing device through a computer network, at least one of:

a transcript of the at least one of: the visual data and the audio data based on executing a corresponding at least one of: the computing platform and a component of the computing platform, and

a device renderable recommendation in a situational awareness context of the incident based on analysis of at least one of: the visual data, the audio data and the sensed at least one environmental parameter.

4. The method of claim 1, further comprising automatically activating at least one Thermal Imaging Camera (TIC) one of: as part of and in addition to the automatic activation of the at least one camera device based on the determination of the occurrence of the incident.

5. The method of claim 1, comprising determining the occurrence of the incident based on determining, through a processor associated with at least one of: the at least one sensor and the data processing device, that the sensed at least one environmental parameter exceeds a threshold value thereof.

6. The method of claim 1, comprising the at least one component of the safety system being Internet of Things (IoT) enabled.

7. The method of claim 1, comprising sensing a parameter related to access of the at least one component of the safety system as the at least one environmental parameter.

8. A safety system of a structure having a fixed piping system implemented therein to supply breathable air thereacross, comprising:

at least one component related to access of the breathable air within the safety system;

at least one sensor associated with the at least one component; and

a data processing device executing a computing platform thereon to integrate the at least one sensor with the computing platform,

wherein, in accordance with the integration of the at least one sensor with the computing platform,

the at least one sensor senses at least one environmental parameter of the at least one component, wherein the at least one environmental parameter includes pressure, smoke, motion, audio detection, temperature, or ambient light, and

a processor associated with the at least one sensor to:

determine, from the sensing of the at least one environmental parameter, an occurrence of an incident corresponding to the at least one environmental parameter; and

automatically activate at least one camera device in a vicinity of the at least one environmental parameter and on the at least one component of the safety system, responsive to the occurrence of the incident.

9. The safety system of claim 8, wherein the automatically activated at least one camera device captures at least one of: visual data and audio data of the incident.

10. The safety system of claim 9, wherein at least one of: the data processing device and another data processing device communicatively coupled to the data processing device through a computer network generates at least one of:

a transcript of the at least one of: the visual data and the audio data based on executing a corresponding at least one of: the computing platform and a component of the computing platform, and

a device renderable recommendation in a situational awareness context of the incident based on analysis of at least one of: the visual data, the audio data and the sensed at least one environmental parameter.

11. The safety system of claim 8, wherein at least one of: the processor associated with the at least one sensor is related to one of: the at least one component and the data processing device, and

wherein the processor automatically activates at least one TIC one of: as part of and in addition to the automatic activation of the at least one camera device based on the determination of the occurrence of the incident.

12. The safety system of claim 8, wherein the processor associated with the at least one sensor determines the occurrence of the incident based on determining that the sensed at least one environmental parameter exceeds a threshold value thereof.

13. The safety system of claim 8, wherein the at least one component is IoT enabled.

14. The safety system of claim 8, wherein the at least one sensor senses a parameter related to access of the at least one component of the safety system as the at least one environmental parameter.

15. A method of a safety system of a structure having a fixed piping system implemented therein to supply breathable air thereacross, comprising:

integrating at least one sensor associated with at least one component of the safety system with a computing platform executing on a data processing device, the at least one component related to access of the breathable air within the safety system;

in accordance with the integration of the at least one sensor with the computing platform,

sensing at least one environmental parameter of the at least one component of the safety system, wherein the at least one environmental parameter includes pressure, smoke, motion, audio detection, temperature, or ambient light;

determining, from the sensing of the at least one environmental parameter, an occurrence of an incident corresponding to the at least one environmental parameter, and

automatically activating at least one camera device in a vicinity of the at least one environmental parameter and on the at least one component of the safety system responsive to the occurrence of the incident; and

in accordance with the automatic activation of the at least one camera device, capturing at least one of: visual data and audio data of the incident.

16. The method of claim 15, further comprising generating, through at least one of: the data processing device and another data processing device communicatively coupled to the data processing device through a computer network, at least one of:

a transcript of the at least one of: the visual data and the audio data based on executing a corresponding at least one of: the computing platform and a component of the computing platform, and

a device renderable recommendation in a situational awareness context of the incident based on analysis of at least one of: the visual data, the audio data and the sensed at least one environmental parameter.

17. The method of claim 15, further comprising automatically activating at least one TIC one of: as part of and in addition to the automatic activation of the at least one camera device based on the determination of the occurrence of the incident.

18. The method of claim 15, comprising determining the occurrence of the incident based on determining, through a processor associated with at least one of: the at least one

sensor and the data processing device, that the sensed at least one environmental parameter exceeds a threshold value thereof.

19. The method of claim **15**, comprising the at least one component of the safety system being IoT enabled.

20. The method of claim **15**, comprising sensing a parameter related to access of the at least one component of the safety system as the at least one environmental parameter.

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